



UNIVERSITY OF CAPE TOWN :: FACULTY OF COMMERCE

Department of Finance and Tax



FTX3045S, FINANCE IIB

Test 1: 29th August 2024

75 Marks

100 Minutes

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. **PLEASE DO NOT TURN THIS PAGE OVER UNTIL INSTRUCTED TO DO SO.**
2. Ensure that your **name, student number, tutor's name and tutorial group number** appears on the cover pages of ALL answer booklets.
3. All **cell phones must be switched off** for the duration of the examination and placed with your other possessions out of sight underneath your chair.
4. You must **remain seated until every paper** has been **collected**.
5. The **invigilators** will **answer no questions relating to the paper** during the test. **Make whatever assumptions you deem appropriate and clearly state these in your answer.**
6. The duration of this examination paper as indicated **includes** reading time.
7. The paper comprises 9 questions. You can **answer all questions in one book.**
8. Write your words in full and **avoid using symbols.** Acronyms should be defined on each question.
9. **Show all your workings. No marks will be awarded for answers that are not backed up with appropriate workings**

This test paper consists of eight pages including the cover page. The **formula sheet** and **calendar** are found at the end of the paper.

QUESTION 1

[27 marks]

The image below displays a snapshot of a bond issued by the South African government. An enlarged image of the snapshot is available on page 8.

11.034/10.994

Settings		Notes		Buy	Sell
29 Bond Description		20 Issuer Description			
Pages	Issuer Information		Identifiers		
11 Bond Info	Name REPUBLIC OF SOUTH AFRICA		FIGI	BBG0000CX531	
12 Addtl Info	Industry Treasury (BCLASS)		ISIN	ZAG000030404	
13 Reg/Tax	Security Information		ID Number	EF5565850	
14 Covenants	Mkt Iss DOMESTIC		Bond Ratings		
15 Guarantors	Ctry/Reg ZA	Currency ZAR	Moody's	Ba2	
16 Bond Ratings	Rank Sr Unsecured	Series R209	S&P	BB	
17 Identifiers	Coupon 6.250000	Type Fixed	Fitch	BB-	
18 Exchanges	Cpn Freq S/A		Composite	BB-	
19 Inv Parties	Day Cnt ACT/365	Iss Price 77.84600	Issuance & Trading		
20 Fees, Restrict	Maturity 03/31/2036		Amt Issued/Outstanding		
21 Schedules	BULLET		ZAR	104,262,552.15 (M) /	
22 Coupons	Iss Yield 8.260		ZAR	104,262,552.15 (M)	
23 Impact	Calc Type		Min Piece/Increment		
Quick Links	Auction Date	07/12/2006	1.00 / 1.00		
30 ALLQ Pricing	Interest Accrual Date	03/31/2006	Par Amount	1.00	
31 QRD Qtr Recap	1st Settle Date	07/21/2006	Book Runner	N/A	
34 TDH Trade Hist	1st Coupon Date	09/30/2006	Exchange	JSE Contrib Prx	
35 CACS Corp Action	AVG YLD=8.26%				
36 CF Filings					
37 CN Sec News					
38 HDS Holders					
44 Send Bond					

Required

RCY
Calculate the realised compound yield for this bond assuming you purchase it at the quoted yield-to-maturity and the settlement date is today. Assume you sell the bond on March 31, 2026, with a settlement date on the same day, at a yield-to-maturity of 6.25%. Use a face value of R10,000 and assume you will realise the following interest rates if you invest in these years: 9% in 2024, 8% in 2025, 9% in 2026, and 3% in 2027.

y₁

y₂

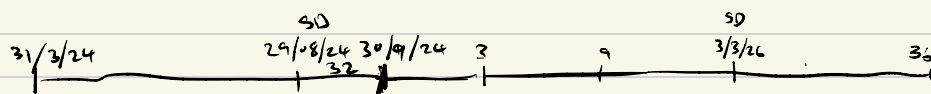
y₃

y₄

[27]

$$RCY = \left[\frac{P_1 - P_0 + \text{coupons} + \text{reinvested inc}}{P_0} \right]^{\frac{1}{n}}$$

$$RCY = \left[\frac{P_1 + \text{coupons} + \text{In from reinvested } C}{P_0} \right]^{\frac{1}{n}} - 1$$



$$FV = 10k = P_1$$

$$\hat{C} = 6,25 \div 2 \times 10k = R312,50$$

$$y = 10,994 \div 2 = 5,497\%$$

$$P_B = 312,50 + 312,5 \times \frac{1 - (1 + 5,497\%)^{-3}}{5,497\%} + \frac{FV}{(1 + 5,497\%)^3}$$

$$= 7.257,70$$

$$AIP = \frac{7.257,7}{(1 + 5,497\%)^{\frac{22}{183}}} = 7.190,10$$

$$AI = 6,25\% \times 10k \times \frac{181}{365} = 258,56$$

$$CP = 7.190,10 - 258,56 = 6.931,54 = P_0$$

$$RCY = \left[\frac{6.931,54 + 10.000}{10.000} \right]^{\frac{1}{n}} - 1$$

selling $p = 10k$ (YTM = Cr)

coupons received: sept 24, mar 25, sept 25 mar 26

coupons reinvested: 312,5

QUESTION 2**[4 Marks]**

The table below presents details of two bonds, Bond A and Bond B. Both bonds have a face value of R1,000 and a credit rating of BB-.

	Issuer	Maturity	Coupon rate	Call price	Call style
Bond A	Sappita	10-years	8%	None	None
Bond B	Sappita	10-years	8%	R1,100	European

Required

Is it true that when interest rates exceed 40% (or are significantly higher than the coupon rate), these two bonds will exhibit similar characteristics? Justify your answer. If true, identify or state the two characteristics that these bonds will share when interest rates are substantially higher than the coupon rates. If false, state two ways in which they will differ. **[4]**

QUESTION 3**[5 Marks]**

Specify two ways you can apply your knowledge of credit ratings to generate returns or to maximise returns on the bond market; furthermore, explain one way you can generate trading profit from one of these. **[5]**

QUESTION 4**[8 Marks]**

Assume the repo rate is currently at 4%, while inflation has risen above 9%, exceeding the target band of 3%-6%, and is expected to continue increasing.

1. Given this information, would you expect the South African Reserve Bank to increase (hike), decrease (cut) or keep the repurchase rate (repo rate) unchanged? Justify your answer and explain how the chosen stance would achieve the bank's objectives **[5]**
2. Based on your expectation from the previous question, explain how you would position your bond portfolio to generate trading profits. **[3]**

QUESTION 5**[4 Marks]**

On July 18, 2024, the South African Reserve Bank (SARB) held its monetary policy meeting and decided to keep the repurchase rate (repo rate) unchanged at 8.25%. Justify this decision by explaining the rationale or the reason behind (i) not increasing and (ii) not reducing the repo rate. **[4]**

QUESTION 6**[6 Marks]**

1. Explain one reason why bonds generally tend to generate higher returns than equities during recessionary periods **[3]**
2. Specify one similarity and one difference between valuing a bond on the coupon date and valuing it between coupon dates. **[3]**

QUESTION 7**[6 Marks]**

1. Identify the two most common shapes of the yield curve and specify the economic condition each predicts. **[2]**
2. On what basis is the yield curve used to predict economic conditions? **[2]**
3. Select one of the curves from the previous question and explain one possible reason for its shape. **[2]**

QUESTION 8**[4 Marks]**

1. Define the term structure of interest rates. **[1]**
2. Explain why the term structure of interest rates is constructed from yields-to-maturities of zero-coupon bonds as opposed to yields-to-maturities of coupon bonds. **[3]**

QUESTION 9**[11 Marks]**

1. State two properties of price risk **[2]**

2. You are given a bond with a coupon of 0%, a face value of R1,000, a time to maturity of 10-years and a yield-to-maturity of 9%. Calculate the following measures for this bond
 - a. Modified duration. **[2]**
 - b. Money duration. **[3]**
 - c. Price value of a basis point. **[2]**
 - d. Assume interest rates increase by 2.3%, estimate the percentage price change for this bond using the modified duration estimate. **[2]**

END OF TEST

FTX3045S Finance IIB-Formula Sheet 1

$$D = \frac{\left(\sum_{t=1}^T PV(CF_t) \times t \right)}{P_0}$$

$$D_{perpetuity} = \frac{(1+y)}{y}$$

$$D_{annuity} = \frac{1+y}{y} - \frac{T}{(1+y)^T - 1}$$

$$D = \frac{1+y}{y} - \frac{(1+y) + T(c-y)}{c[(1+y)^T - 1] + y}$$

$$D_{parbond} = \frac{1+y}{y} \left[1 - \frac{1}{(1+y)^T} \right]$$

$$FVIFA_{ordinary} = \frac{[(1+r)^n - 1]}{r}$$

$$FVIFA_{due} = \frac{[(1+r)^n - 1] \times (1+r)}{r}$$

$$PVIFA_{ordinary} = \frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r}$$

$$PVIFA_{due} = \left[\frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r} \right] \times (1+r)$$

$$Convexity = \frac{(PV_-) + (PV_+) - 2 \times (PV_0)}{(\Delta yield)^2 \times PV_0}$$

$$D^* = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Yield) \times (PV_0)}$$

Calendar for Year 2024 (South Africa)

January Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 4:☉ 11:☿ 18:☿ 25:☉	February Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 3:☉ 10:☿ 16:☿ 24:☉	March Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 3:☉ 10:☿ 17:☿ 25:☉
April Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 2:☉ 8:☿ 15:☿ 24:☉	May Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 1:☉ 8:☿ 15:☿ 23:☉ 30:☉	June Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 6:☿ 14:☉ 22:☉ 28:☿
July Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 6:☿ 14:☉ 21:☉ 28:☿	August Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 4:☿ 12:☉ 19:☉ 26:☿	September Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 3:☿ 11:☉ 18:☉ 24:☿
October Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 2:☿ 10:☉ 17:☉ 24:☿	November Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 1:☿ 9:☉ 15:☉ 23:☿	December Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 1:☿ 8:☿ 15:☉ 23:☿ 31:☿
Holidays: 1 Jan New Year's Day 21 Mar Human Rights Day 29 Mar Good Friday 1 Apr Family Day 27 Apr Freedom Day 1 May Workers' Day 16 Jun Youth Day 17 Jun 'Youth Day' day off 9 Aug National Women's Day 24 Sep Heritage Day 16 Dec Day of Reconciliation 25 Dec Christmas Day 26 Dec Day of Goodwill		

11.034/10.994

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99 Buy

99 Sell

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Issuer Information

Name REPUBLIC OF SOUTH AFRICA

Industry Treasury (BCLASS)

Security Information

Mkt Iss DOMESTIC

Ctry/Reg ZA Currency ZAR

Rank Sr Unsecured Series R209

Coupon 6.250000 Type Fixed

Opn Freq S/A

Day Cnt ACT/365 Iss Price 77.84600

Maturity 03/31/2036

BULLET

Iss Yield 8.260

Calc Type (368)S AFRICA EX-DIV BD

Auction Date 07/12/2006

Interest Accrual Date 03/31/2006

1st Settle Date 07/21/2006

1st Coupon Date 09/30/2006

AVG YLD=8.26%.

Identifiers

FIGI BBG0000CX531

ISIN ZAG000030404

ID Number EF5565850

Bond Ratings

Moody's Ba2

S&P BB

Fitch BB-

Composite BB-

Issuance & Trading

Amt Issued/Outstanding

ZAR 104,262,552.15 (M) /

ZAR 104,262,552.15 (M)

Min Piece/Increment

1.00 / 1.00

Par Amount 1.00

Book Runner N/A

Exchange JSE Contrib Prx